Synopsis

Loan Default Prediction Using Machine Learning Techniques

Problem Id: 2

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Loan Default Prediction using Machine Learning

1. Problem Statement

The main aim of this project is to build a machine learning system that can predict whether a customer is likely to default on a loan. This is done using banking data. The process begins with Exploratory Data Analysis (EDA), which helps in understanding the data, identifying patterns, and any issues early on.

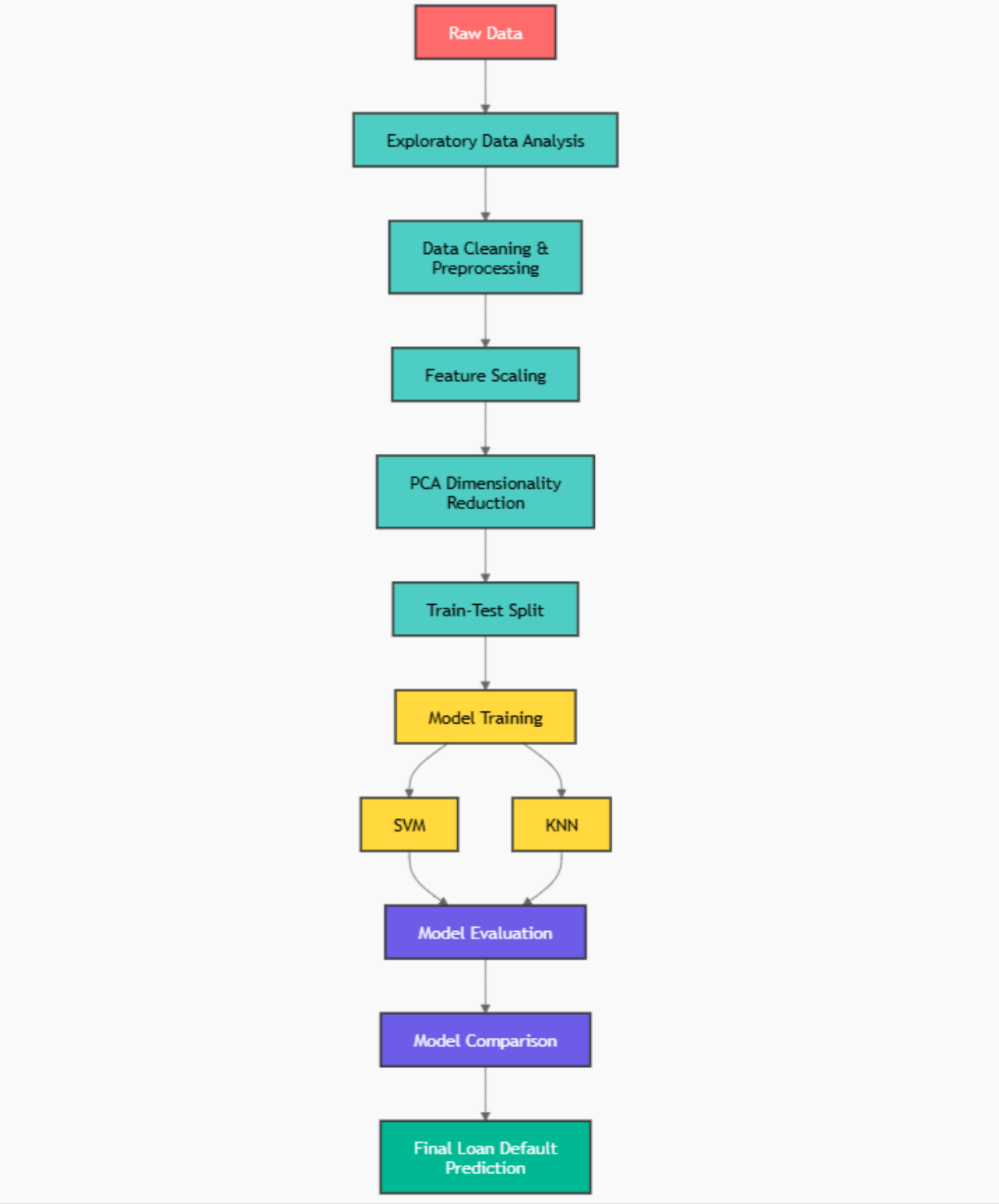
After that, data preprocessing is carried out to handle missing or inconsistent values, normalize numerical features, and prepare the dataset so that the models can perform better.

Since banking datasets usually contain a large number of variables, Principal Component Analysis (PCA) is used to reduce the complexity of the data while keeping the most important information.

The system then trains different models, such as Support Vector Machine (SVM) with various kernel functions (like linear, polynomial, and RBF) and K-Nearest Neighbors (KNN). These models help identify patterns that indicate the risk of a customer defaulting on a loan.

To evaluate how well the models perform, metrics like precision, recall, F1 score, and ROC-AUC are used. These measures give a clear idea of how accurately the system can predict loan defaults.

2.System Workflow Diagram (Fig. 1)



Explanation of Each Step

3.1 Dataset Collection

First, we collect the dataset, which mainly includes details that show how financially stable a person is. For example, income and employment status are very important. If someone has a stable job and earns well, it usually means they can repay the loan without much trouble. On the other hand, if a person has an irregular income or no job, the risk for the bank becomes higher.

3.2 Exploratory Data Analysis (EDA)

After collecting the data, the next step is to analyze the data. This is where EDA comes in.

We use tools like:

- Histograms to see how data is distributed
- Heatmaps to check how different features are related
- Box plots to spot unusual values or outliers

The idea here is not to jump into modeling right away, but to first get comfortable with the data and find patterns, notice anything unusual, and understand which factors might actually matter.

3.3 Data Preprocessing

Before using the data in any model, we need to clean it up. Real-world data is rarely perfect as it often has missing values, inconsistent formats, or non-numeric data.

So in this step, we:

- Fill or remove missing values
- Convert categorical data into numbers
- Scale the features so everything is on a similar range

This step is important because by this machine learning models work much better when the data is clean and well-prepared.

3.4 Dimensionality Reduction (PCA)

Sometimes datasets have too many features, which can make things complicated and slow. To handle this, we use PCA.

In simple terms, PCA reduces the number of variables while keeping the most useful information. It helps remove unnecessary noise and makes the model more efficient.

Because of this, the model trains faster and often performs better overall.

3.5 Model Training

(A) **Support Vector Machine (SVM)**
SVM is a powerful classification algorithm that finds the optimal boundary (hyperplane) between different classes. It works well in high-dimensional datasets and can handle complex relationships using kernel functions such as linear and RBF kernels.

(B) **K-Nearest Neighbors (KNN)**
KNN is a distance-based algorithm that classifies data based on the majority class of its nearest neighbors. It is simple and easy to implement but its performance depends on the choice of neighbors and feature scaling

4. Why These Models Are Used?

Support Vector Machine (SVM):
Support Vector Machine is chosen because it performs very well in classification problems where the dataset contains multiple features and complex relationships between variables. SVM works by finding the optimal hyperplane that separates different classes in the dataset with the maximum margin. This helps the model generalize well on unseen data.

KNN (K-Nearest Neighbors) is used for classification based on similarity between data points.
It does not build a model; instead, it stores all training data.
For a new input, it finds the nearest k data points using distance.

The class is decided by majority vote of these neighbors.
It is simple, easy to understand, and works well

5. Why Other Models Are Not Preferred?

Some other models were considered, but they have certain drawbacks in this case:

- **Decision Trees** can easily overfit, especially with complex data
- **Random Forest** gives good results but is more harder to interpret.
- **Neural Networks** are powerful but require large datasets.

6. What If We Used Other Models?

When building a system to predict loan defaults, we looked at other machine learning models as well. Here's why they might not work as well for this project:

1. Decision Trees
Decision Trees can learn the training data very closely, which can cause **overfitting**. This means they may perform well on the data they have seen before but may make mistakes on new data.

2. Random Forest
Random Forest combines many Decision Trees, which usually improves accuracy and reduces overfitting. However, it is **harder to understand and explain**. For financial decisions, being able to explain why a prediction was made is very important.

3. Neural Networks
Neural Networks are very powerful and can learn complex patterns. But they need **a lot of data and computing power**, and they are **difficult to interpret**.

This makes them less practical for smaller datasets or situations where explanations are important.

7. Model Evaluation Metrics

Accuracy

Measures overall correctness of predictions.

$$\boxed{\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}}$$

Precision

Measures how many predicted defaulters are actually defaulters.

$$\boxed{\text{Precision} = \frac{TP}{TP + FP}}$$

Recall

Measures how many actual defaulters are correctly identified.

$$\boxed{\text{Recall} = \frac{TP}{TP + FN}}$$

F1-Score

Harmonic mean of precision and recall.

$$\text{F1 Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

ROC-AUC

Measures model's ability to distinguish between classes.

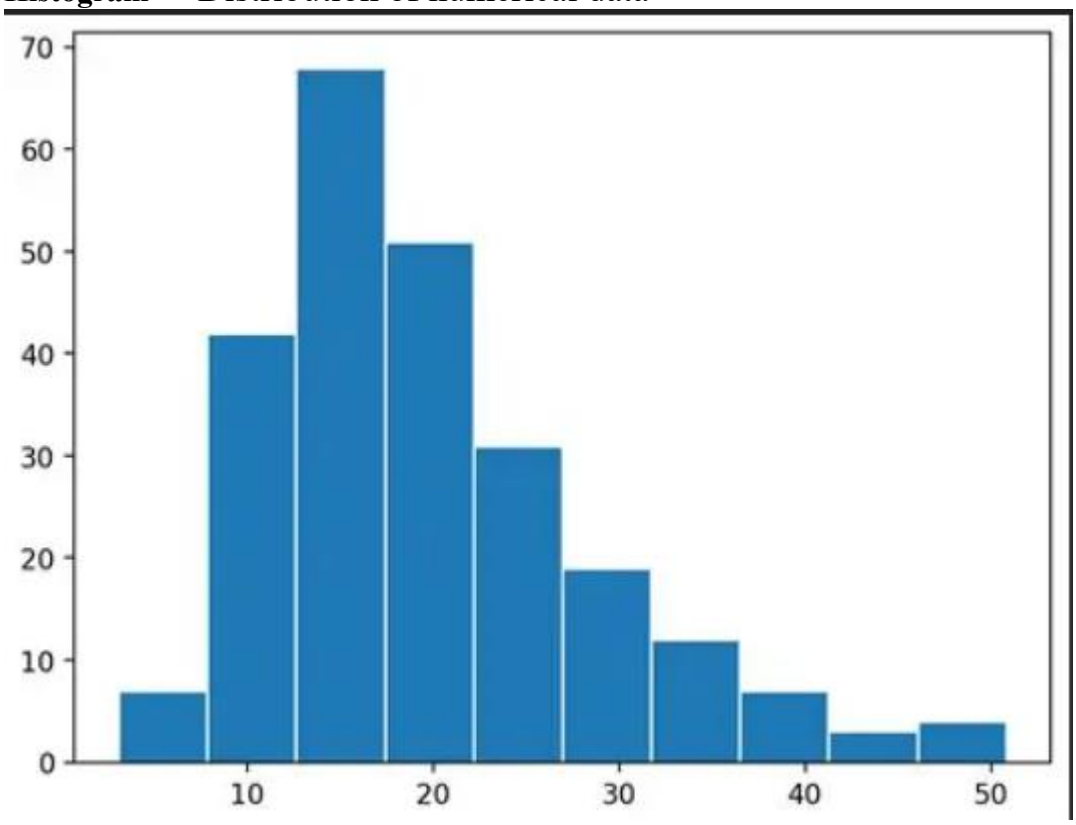
True Positive Rate (TPR) vs False Positive Rate (FPR)

$$TPR = \frac{TP}{TP + FN}$$

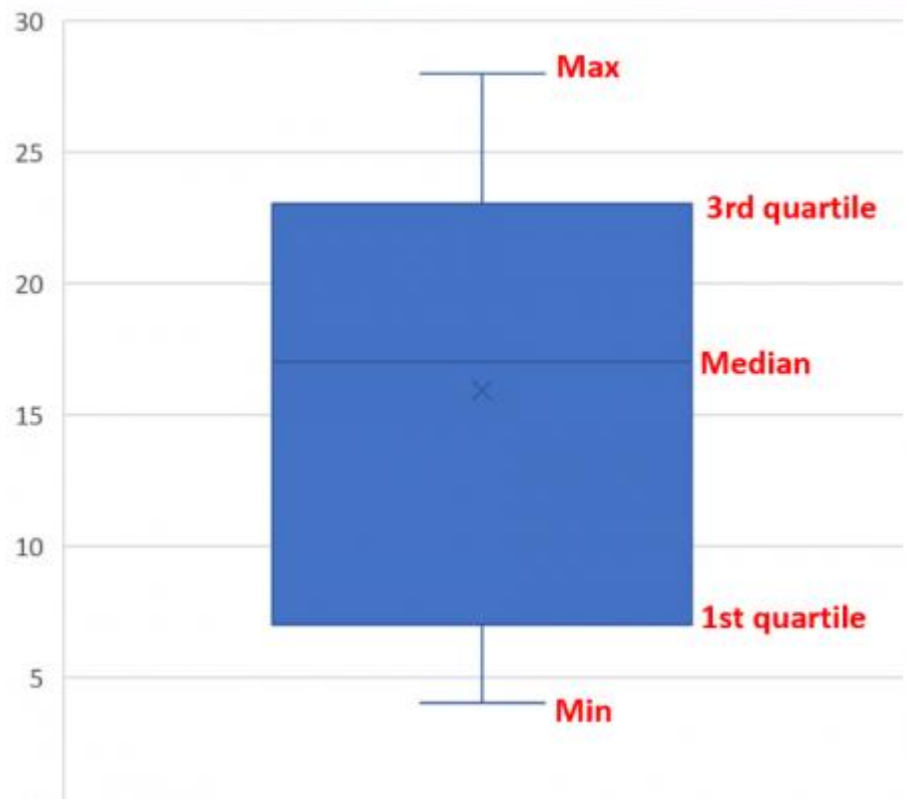
$$FPR = \frac{FP}{FP + TN}$$

8. Visualization Techniques

- **Histogram** → Distribution of numerical data



- **Box Plot** → Outlier detection



- **Heatmap** → Feature correlation

9. Libraries Used

In this project, we use some important Python tools, called libraries, to help us work with data and build the machine learning system.

1. NumPy
NumPy helps us work with numbers and math very quickly. For example, we can add, subtract, or do other calculations on big lists of numbers, like customers' incomes or loan amounts. It's fast and very useful when dealing with large datasets.

2. Pandas
Pandas is like Excel inside Python. It helps us store, clean, and manage data easily. We can fill missing values, change words into numbers and filter out unnecessary information. Pandas makes the data ready to use for analysis and machine learning.

3. Matplotlib
Matplotlib helps us draw basic graphs like line charts, bar charts, and histograms. These graphs help us see patterns in the data.

4. Seaborn → Beautiful and Clear Graphs
Seaborn is like Matplotlib, but prettier and more colorful. It makes it easy to create graphs.

5. Scikit-learn → Machine Learning Models
Scikit-learn is the library we use to teach the computer how to make predictions.

10. Required Topics

1. Exploratory Data Analysis (EDA)

EDA is basically the first step where we try to "get to know" the dataset before doing anything else. Instead of directly applying machine learning, we first explore the data to understand what it looks like.

We check things like how the data is distributed, whether there are any unusual values, and how different variables are related to each other. This is usually done using graphs like histograms, heatmaps, and box plots.

The main idea of EDA is simple: understand the data well so that we can make better decisions later while building the model.

2. Data Preprocessing

Raw data is almost never clean or ready to use. That's where data preprocessing comes in. It helps convert messy, incomplete data into a format that machine learning models can actually work with.

In this step, we handle missing values, fix inconsistencies, and convert categorical data into numerical form. We also scale the data so that all features are on a similar range.

This step is very important because even a good model won't perform well if the input data is poor. Clean data = better results.

3. Dimensionality Reduction (PCA)

Sometimes datasets have too many features, which can make the model slow and less efficient. PCA helps solve this problem by reducing the number of features while still keeping the most important information.

4. Support Vector Machine (SVM)

SVM is a powerful classification algorithm that tries to separate data into different classes using the best possible boundary.

In simple terms, it draws a line between two groupS.in this case, people who will repay the loan and those who might default. The goal is to place this boundary in such a way that it clearly separates both groups.

SVM is especially useful when the data is complex because it can handle both simple (linear) and complicated (non-linear) patterns using different kernel functions.

5. K-Nearest Neighbors (KNN)

KNN is one of the simplest machine learning algorithms. It works on a very basic idea: similar things are usually close to each other.

When a new data point comes in, KNN looks at its nearest neighbors (the most similar existing data points) and assigns a class based on the majority.

For example, if most of the nearby customers are defaulters, the new customer will also likely be classified as a defaulter.

It’s easy to understand and works well for smaller datasets, but it can become slow when the dataset is very large.

11. Platform Used

- Jupyter Notebook /
  Visual Studio Code /Python IDLE

12. Conclusion

This system helps financial institutions:

- Predict loan defaults accurately
- Reduce financial risk
- Make data-driven decisions

By combining preprocessing, PCA, and powerful models like SVM and K-NN, the system achieves efficient and reliable predictions.

13. References

- Scikit-learn Documentation: https://scikit-learn.org
- https://scikit-learn.org/stable/modules/preprocessing.html
- https://arxiv.org/abs/1907.01800